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MODEL SELECTION AND VALIDATION REPORT ( general_kde_fric model )
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## OBSERVATIONS:3451

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( general_kde_fric model )

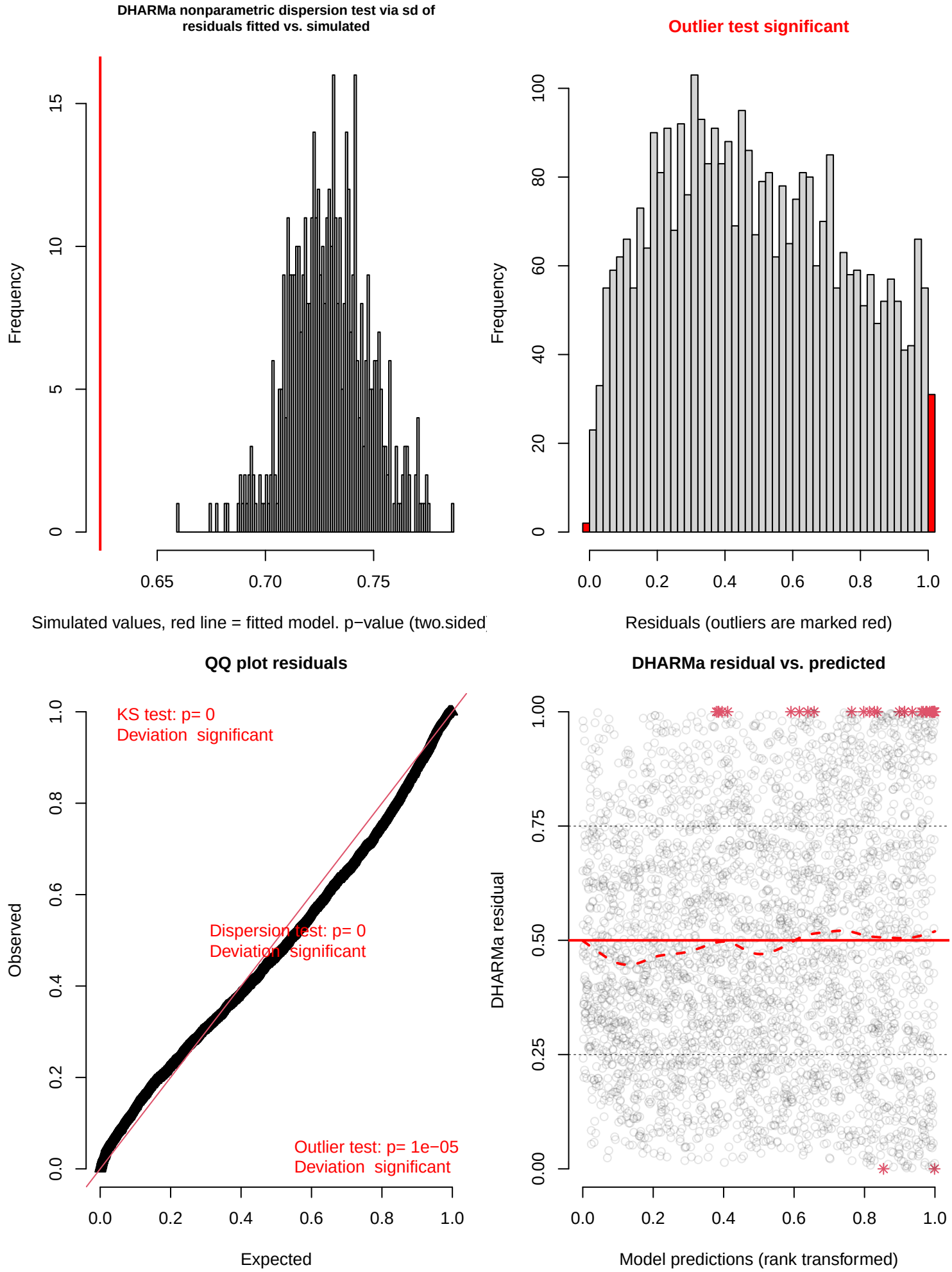
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

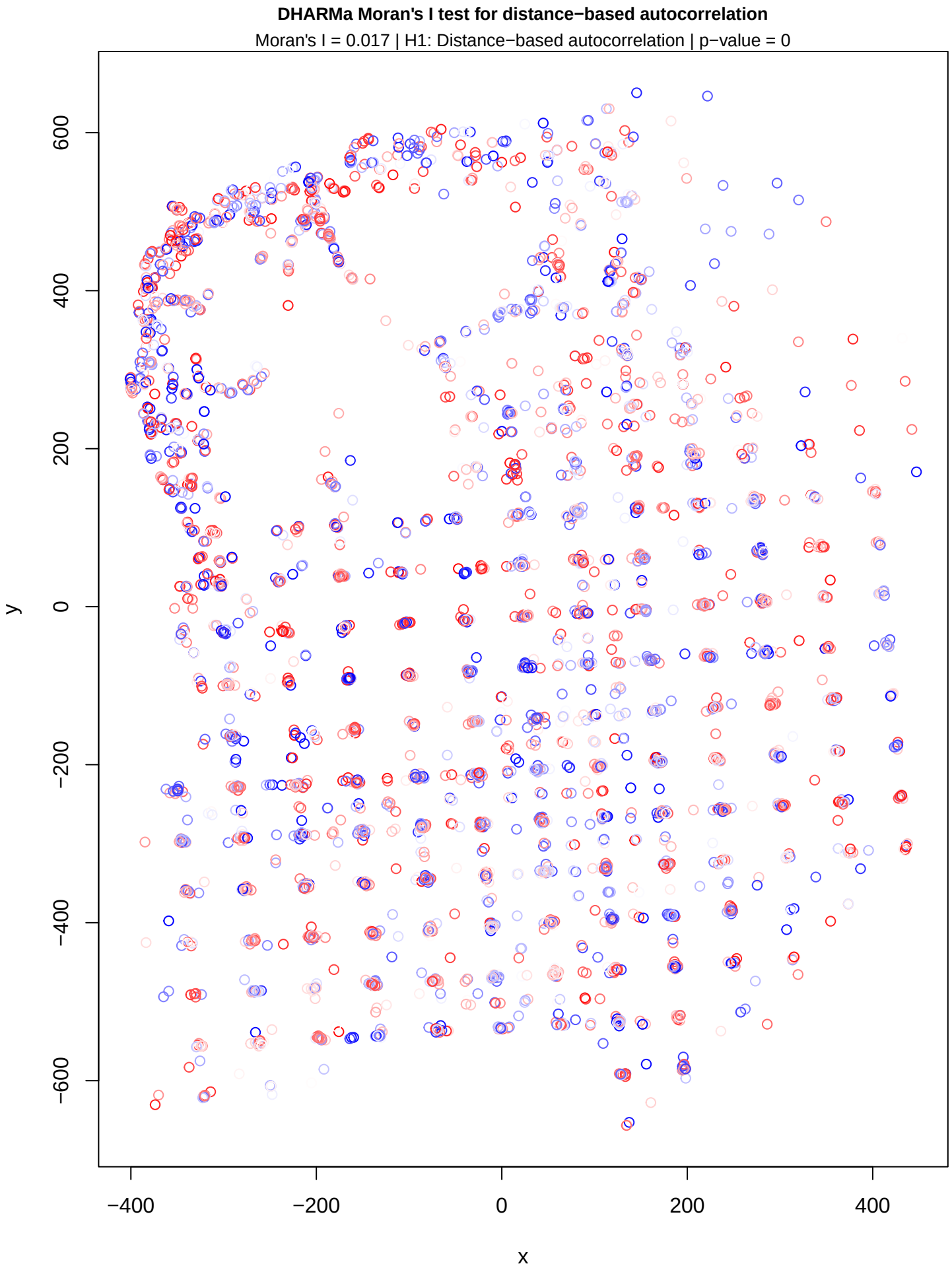
Conditional model:
              coef.est coef.se
(Intercept)      −0.09    0.13
sbt_mean          0.12    0.05
sbt_sd            0.04    0.03
sst_sd            0.05    0.02
sic_mean         −0.03    0.04
log_chla_mean     0.03    0.02
depth            −0.23    0.03

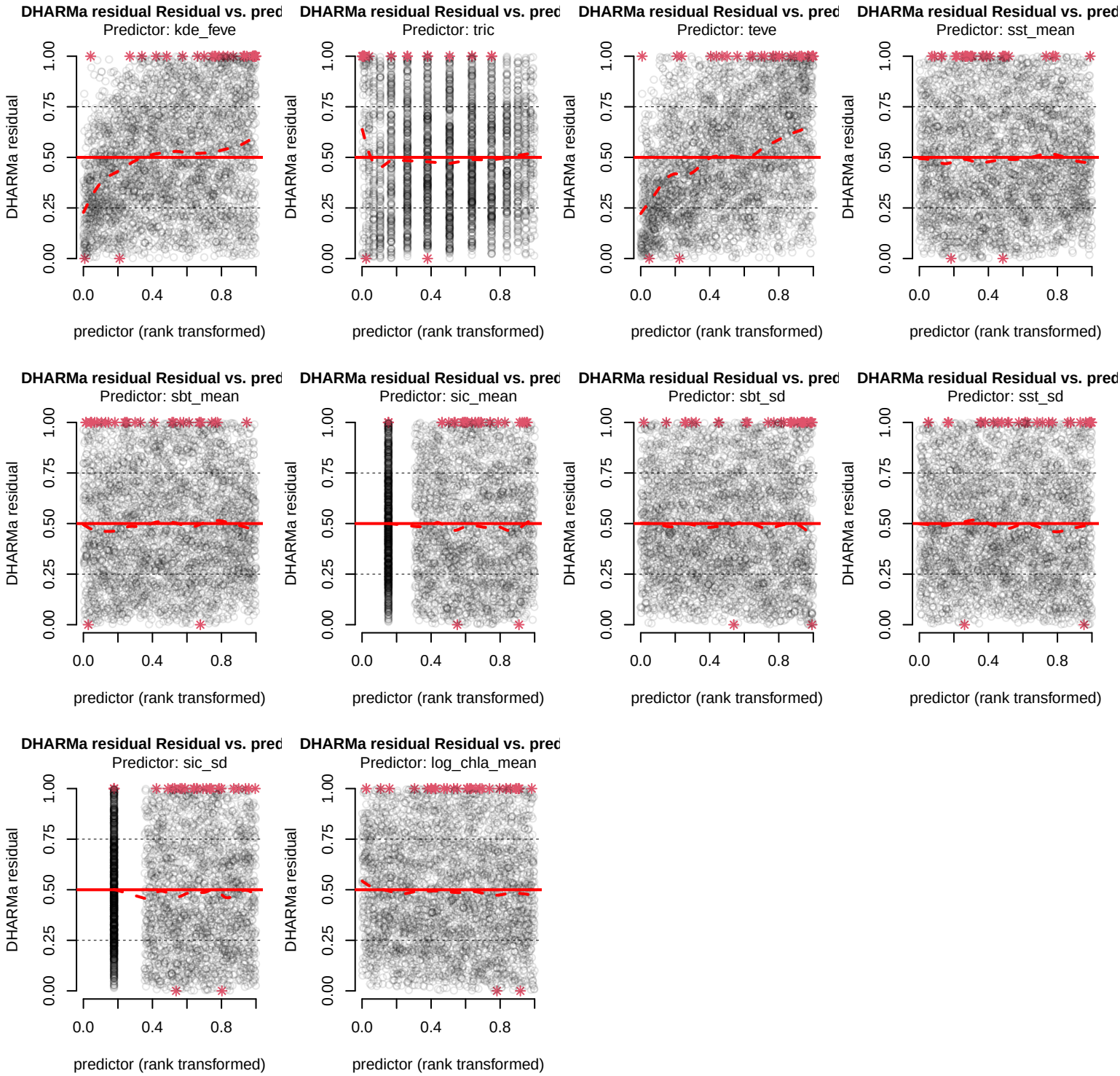
Dispersion parameter: 0.83
Spatiotemporal AR1 correlation (rho): 0.94
Matérn range: 235.12
Spatiotemporal marginal AR1 SD: 0.59
REML criterion at convergence: 4476.371

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                   11 8974.742   1.571195
spatiotemporal_random_fields_ar1_with_anisotropy 13 8976.313  80.869207
spatial_random_fields                              10 9057.182   2.105615
spatial_random_fields_with_anisotropy              12 9059.288 206.379524
spatiotemporal_random_fields_iid_with_anisotropy 12 9265.667   3.432451
spatiotemporal_random_fields_iid                   10 9269.100 268.812740
error_only                                          8 9537.913         NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
0.65
0.70
0.75
Simulated values, red line = fitted model. p−value (two.sided
Outlier test significant
Frequency
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0
Deviation significant
Dispersion test: p= 0
Deviation significant
Outlier test: p= 1e−05
Deviation significant
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected
DHARMa residual vs. predicted
DHARMa residual
0.00
0.25
0.50
0.75
1.00
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.017 | H1: Distance−based autocorrelation | p−value = 0
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: kde_feve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8